
1. 📌 Project Overview

The system will manage:

- Student onboarding
- Multi-sport batch management
- Complex fee structures (Regular / Alternate / Day-based)
- Subscription lifecycle
- Flexible time-slot handling
- Staff & payroll
- Attendance tracking
- WhatsApp notifications

The system must be dynamic, configuration-driven, and scalable.

2. 👤 Student Management

2.1 Student Onboarding

System must allow:

- Add new student
- Capture parent details
- Assign sport
- Assign level
- Assign plan type (Regular / Alternate / Day-based)
- Assign duration (1 / 3 / 6 months)
- Assign time slot
- Generate subscription start & expiry
- Upload documents
- Store photo

Fields:

- Student ID (Auto-generated)
 - Name
 - DOB
 - Gender
 - Contact Number
 - Parent Name
 - Parent WhatsApp Number
 - Address
 - Sport
 - Level
 - Plan Type
 - Duration
 - Assigned Time Slot
 - Admission Date
 - Subscription Start Date
 - Expiry Date
 - Status (Active / Expired / Suspended)
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3. 🎮 Sports & Fee Configuration

System must support 5 Sports:

1. Badminton
2. Tennis
3. Football
4. Skating
5. Chess

3.1 Level Structure

Each sport can have:

- Beginner
- Intermediate
- Adults
- Advance
- All Players (Football)
- Day-based model (Chess)

System must allow:

- Level-specific duration hours
 - Enable/Disable alternate plan per level
 - Different pricing per duration
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3.2 Plan Types

Regular Plan

Full weekly schedule

Alternate Plan

Reduced frequency
(Not available for certain levels)

Day-Based Plan (Chess)

- 6 Days
 - 3 Days
 - 2 Days
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3.3 Duration Options

- 1 Month
- 3 Months
- 6 Months

Each duration:

- Has independent pricing
 - Not auto-calculated unless enabled
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3.4 Pricing Configuration

Admin must be able to:

- Add/edit pricing
- Set effective date
- Maintain price history
- Disable specific plan
- Configure pro-rata (optional)
- Set discount %

System must NOT hard-code pricing.

4. 🕒 Time Slot & Batch Management

4.1 Time Slot Configuration

Per sport:

- Multiple time slots
- Start time
- End time
- Capacity limit

System must:

- Prevent overbooking

- Show live occupancy
 - Allow admin override
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4.2 Batch Assignment

Each student must be assigned:

- Sport
 - Level
 - Plan type
 - Time slot
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5. Special Case: Flexible Slot Switching

5.1 Temporary Slot Change

If a student assigned to 4–5 PM wants to attend 1–2 PM:

System must allow:

- Attendance in new slot
- Track actual attended slot
- Keep original subscription unchanged

Admin must:

- Approve slot override
- Set date range
- Record reason

Auto revert after end date.

5.2 Permanent Slot Change

Admin can:

- Change assigned slot

- Update capacity
 - Maintain slot history
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5.3 Attendance Logic

Attendance must store:

- Student ID
- Date
- Assigned Slot
- Attended Slot
- Override flag

System validations:

- Subscription must be active
 - Sport must match
 - Slot capacity check
 - Admin override allowed
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6. Attendance Management

System must allow:

- Daily attendance marking
- Slot-wise attendance
- Batch-wise attendance
- Monthly reports
- Attendance %

Reports:

- Student attendance report
- Slot utilization report

- Override attendance report
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7. Subscription & Payment Management

7.1 Subscription Rules

- No refund
- No extension
- Renewal starts from expiry date
- Early renewal allowed but effective from expiry

System must display on invoice:

"NO REFUND AND NO EXTENSION FOR ALL ACTIVITIES."

7.2 Mid-Month Joining

System must support:

Option A: Full charge

Option B: Pro-rata (Admin configurable)

7.3 Renewal Management

System must:

- Track renewal history
 - Generate invoices
 - Track payment mode
 - Track pending dues
 - Send expiry reminders
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8. Payment Tracking

Fields:

- Invoice ID
- Student ID
- Amount
- Payment Mode (Cash / UPI / Online)
- Transaction ID
- Payment Date
- Status (Paid / Partial / Pending)

Reports:

- Monthly revenue
 - Sport-wise revenue
 - Pending dues
 - Renewal rate
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9. 🏠 Staff & Payroll Management

9.1 Staff Onboarding

- Add staff
- Assign role
- Assign batches
- Create login

Roles:

- Super Admin
 - Admin
 - Trainer
 - Accountant (Staff)
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9.2 Salary Management

System must support:

- Fixed salary
- Per batch salary
- Salary revision history
- Payroll processing
- Payslip generation

Fields:

- Old salary
- Revised salary
- Effective date
- Reason

10. 📡 Communication & Notification System

10.1 Parent Onboarding

- Link parent with student
- Store WhatsApp number
- Support multiple students under one parent

10.2 WhatsApp Automation

System must support:

- Fee due reminder
- Expiry reminder (7 days before)
- Expiry reminder (3 days before)
- Expiry day reminder
- Custom message
- Holiday notice

- Batch timing change

Must support dynamic variables:

- {StudentName}
- {ParentName}
- {Sport}
- {ExpiryDate}
- {DueAmount}
- {TimeSlot}

Message log must be stored.

11. 🏆 Dashboard & Reporting

Admin Dashboard

- Total students
 - Active subscriptions
 - Expiring in 7 days
 - Revenue summary
 - Pending payments
 - Slot occupancy
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Trainer Dashboard

- Assigned slots
 - Daily attendance
 - Student list
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Reports

- Sport-wise revenue

- Level-wise revenue
 - Regular vs Alternate ratio
 - Duration distribution
 - Attendance %
 - Slot utilization
 - Slot override tracking
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12. Security & Access Control

Role-based access control required.

Permissions:

- Student CRUD
- Fee edit
- Payroll edit
- Attendance marking
- Notification sending
- Report view

Passwords must be encrypted.

13. Non-Functional Requirements

Performance:

- Support 1000+ active students
- Dashboard load < 3 seconds

Security:

- Role-based restriction
- Secure payment logs

Backup:

- Daily automated backup

Deployment:

- Web-based (Cloud hosted)
 - Mobile responsive – Attendance view
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14. Database Structure (High Level)

Required core tables:

- students
- parents
- sports
- sport_levels
- fee_plans
- time_slots
- slot_overrides
- subscriptions
- payments
- attendance
- staff
- payroll
- notifications

All pricing must be configuration-driven.

15. Future Enhancements (Optional Phase 2)

- Online payment gateway
- QR attendance
- Mobile app

- Biometric attendance
 - Certificate generation
 - Performance tracking
 - Multi-branch support
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Final Summary

This system must support:

- ✓ Complex sport-wise pricing
 - ✓ Regular / Alternate / Day-based plans
 - ✓ Flexible slot switching
 - ✓ Capacity management
 - ✓ Automated reminders
 - ✓ Subscription lifecycle
 - ✓ Payroll tracking
 - ✓ Advanced reporting
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